

PAPER • OPEN ACCESS

Detection of ADHD from EEG signals using new hybrid decomposition and deep learning techniques

To cite this article: Mustafa Yasin Esas and Fatma Latifoğlu 2023 *J. Neural Eng.* **20** 036028

View the [article online](#) for updates and enhancements.

You may also like

- [Classification of attention deficit/hyperactivity disorder based on EEG signals using a EEG-Transformer model](#)
Yuchao He, Xin Wang, Zijian Yang et al.
- [Neurological state changes indicative of ADHD in children learned via EEG-based LSTM networks](#)
Yang Chang, Cory Stevenson, I-Chun Chen et al.
- [Use of deep learning to detect personalized spatial-frequency abnormalities in EEGs of children with ADHD](#)
He Chen, Yan Song and Xiaoli Li

OPEN ACCESS



CrossMark

RECEIVED

13 September 2022

REVISED

17 March 2023

ACCEPTED FOR PUBLICATION

30 March 2023

PUBLISHED

5 June 2023

MADE OPEN ACCESS

25 September 2023

Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](#).

Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.



PAPER

Detection of ADHD from EEG signals using new hybrid decomposition and deep learning techniques

Mustafa Yasin Esas^{ID} and Fatma Latifoğlu^{* ID}

Department of Biomedical Engineering, Erciyes University, Kayseri, Türkiye

^{*} Author to whom any correspondence should be addressed.E-mail: flatifoglu@erciyes.edu.tr and flatifoglu38@gmail.com**Keywords:** ADHD, EEG, deep learning, local mean decomposition, variational mode decomposition, classification

Abstract

Objective. Attention deficit hyperactivity disorder (ADHD) is considered one of the most common psychiatric disorders in childhood. The incidence of this disease in the community draws an increasing graph from the past to the present. While the ADHD diagnosis is basically made with the psychiatric tests, there is no active clinically used objective diagnostic tool. However, some studies in the literature has reported development of an objective diagnostic tool that facilitates the diagnosis of ADHD. **Approach.** In this study, it was aimed to develop an objective diagnostic tool for ADHD using electroencephalography (EEG) signals. In the proposed method, EEG signals were decomposed into subbands by robust local mode decomposition and variational mode decomposition techniques. These subbands and the EEG signals were fed as input data to the deep learning algorithm designed in the study. **Main results.** As a result, an algorithm has been put forward that distinguishes over 95% of ADHD and healthy individuals through using a 19-channel EEG signal. In addition, a classification accuracy of over 87% was obtained by the proposed approach of EEG signal decomposition followed by data processing in the designed deep learning algorithm. **Significance.** The findings of the current research enrich the literature based on originality and proposed method can be used as a clinical diagnostic tool in the near future.

1. Introduction

Attention deficit hyperactivity disorder (ADHD) is one of the most common neuropsychiatric disorders in the children today. While an average of 5% in the society [1] deals with the ADHD, its symptoms typically appear before the age of 12 [2, 3]. As for the diagnosis of ADHD, international classification system DSM-V (Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition) is utilized.

ADHD can cause a decrease in the school performance in children and a deterioration in the communication with family and friends. Depending on this disease, risky behaviours such as aggression may develop in children in the future [4]. There is no specific diagnostic tool for ADHD. Clinicians diagnose ADHD based on symptoms such as inattention and hyperactivity. Diagnostic procedures include evaluation of ADHD criteria in DSM-V, medical examination, information from school, interviews with family members, ADHD rating scales such as Conners

Rating, and neuropsychological tests [5, 6]. No commonly used objective assessment tools support the diagnosis and eliminate or minimise subjectivity. There are studies carried out to support the diagnosis of ADHD. However, these studies are not yet in clinical use.

Electroencephalography (EEG) is the process of monitoring and recording electrical signals and voltage fluctuations created by neuron activity in the brain with electrodes attached to a person's scalp. EEG, a non-invasive method, can record brain activity for a long time and measure signals. Therefore, EEG is analysed for treating health problems associated with the brain activity. EEG includes information about the neural activity of a person's brain [7] which is used for research areas such as neuroscience, biomedical engineering and brain-computer interface studies. The cost-effectiveness and non-invasive nature of the EEG makes it an important source of data for many neuroscience studies [8].

Adamou *et al* reviewed the use of EEG signals for diagnosing ADHD in adults and stated that the review publication examined 21 papers. It was concluded that although the use of EEG could assist in diagnosing ADHD, the current state of associated knowledge was still insufficient. Additionally, the theta beta ratio (TBR), which is considered an important marker for ADHD in children, was found to be quite inconsistent in adults [9].

Arns *et al* reported the utility of EEG TBR in ADHD diagnosis by examining 1253 ADHD and 517 healthy children and adolescents aged 6–18 years. It was stated that TBR cannot be accepted as a reliable diagnostic measure of ADHD and can be used as a prognostic measure rather than a diagnostic measure [10]. Another similar study stated that ADHD was correctly detected in 166 out of 185 participants using the TBR parameter. This study stated that TBR has a high ADHD discrimination potential but it is insufficient to substitute clinical interviews alone [11].

Tanko *et al* presented a new feature extraction method based on manually modelled one-dimensional signal classification system using EEG signals to detect ADHD. The proposed method is the learning network model with a directed graph and an eight-pointed star pattern (EPSPat). While the method focused on analysing the abnormalities in the EEG signals, it was claimed by that ADHD was detected with a high accuracy rate [12].

Vahid *et al* applied deep learning methods to event-related EEG data for distinguishing between ADHD and healthy individuals. In addition, ADHD subtypes were tried to be determined using the same deep learning methods. The results showed that the applied deep learning model ('EEGNet') was able to distinguish both ADHD subtypes and healthy controls with up to 83% accuracy [13]. In another study investigating whether EEG can identify ADHD subtypes or not, showed that EEG produces significant results for identifying ADHD but fell short of defining subtypes [14].

Quintana *et al* studied the potential of EEG for detecting the presence of ADHD in a clinical sample. The EEG marker was detected in 15 out of 16 (sensitivity of 94%) participants with ADHD. It was stated that EEG markers were not detected (specificity of 100%) in any of the patients who showed ADHD-like symptoms but did not ADHD. These results indicate that EEG can be used as a marker to detect ADHD [15].

Clarke *et al* analysed the changes in the EEG signals in the participants having ADHD with a gap of 11 years. Twenty-five males were reassessed 11 years after the initial examination which had taken place when they were 8–12 years old. EEG was recorded at rest with eyes closed, and delta, theta, alpha and beta waves were calculated for both periods. According to the findings, significant changes were observed in the EEG activity from childhood to adulthood and

significant developmental changes were found in the subjects' EEGs [16].

An EEG-based deep learning study was examined to identify children with ADHD. Among the methods examined in the study, it was stated that the learning abilities of convolutional neural networks (CNNs) are remarkable [17]. In another study performed on 20 healthy and 20 ADHD adults, the results obtained from the classification using different methods and datasets indicated that CNN achieves high accuracy and significantly outperforms recurrent neural networks (RNNs) [18]. Another study showed that it is possible to reach an accuracy of 90% when using CNN techniques for ADHD identification [19]. In another study, EEG signals from 31 ADHD and 30 healthy children were preprocessed and separated into theta, alpha, beta and gamma frequency bands and an image matrix was formed. It was stated that overall success rates of over 90% were achieved in the classification study using the 13-layer CNN structure [20].

This study used the CNN structure, the superior features of which have been expressed explicitly and routinely in the literature.

In a research study covering the last 75 years on the use of EEG in the diagnosis of ADHD, it was stated that it was not possible to find a simple diagnostic criterion for ADHD. It was stated in the study that the EEG is not ready to be used as a tool to diagnose or assist in the diagnosis of ADHD. However, it has been stated that ADHD detection studies with high (>90%) accuracy rates reported using EEG are much more successful than state-of-the-art classification studies using anatomical and functional MRI data [21].

A study on the effect of spectral properties of EEG signals on ADHD classification was reviewed. It was stated that the study's primary purpose was to reveal the most effective EEG channel and the most influential recording status for the diagnosis of ADHD. This way, it was aimed to make ADHD classification with fewer channels and higher accuracy. According to the findings, it was stated that 88% (eyes open) accuracy was calculated in the 'Fp1, F7' channel. It was seen that the 'Fp1, F7' channel gave the highest success [22]. In another study, differences in information flow in the brain's frontal region were found between individuals with ADHD and healthy individuals [23].

In this study, we performed a classification analysis by detecting the possible changes in the EEG signals of healthy individuals and those with ADHD for the first time in the literature with robust local mode decomposition (RLMD), variational mode decomposition (VMD), and deep learning algorithms.

EEG signals were divided into subbands by RLMD and VMD techniques. The subband data was applied as input data to the deep learning algorithm with the signal itself, and the classification study was performed. Thus, a new model that

achieves high-accuracy classification success has been introduced.

We can summarise the flow of our study as follows: in section 1, information about ADHD and EEG is given. In the literature, summary information is given about the studies on ADHD using EEG and deep neural networks. In the second part, section 2, EEG data, VMD and RLMD processes, deep neural network structure, and designed deep learning algorithm are expressed under the headings. The path followed in our study is indicated. In the third chapter, the results are presented, and the findings are discussed.

2. Materials and methods

In our study, the EEG data obtained from the database (<https://iee-dataport.org/open-access/eeg-data-adhd-control-children>) [24] was applied as an input to the RLMD and VMD algorithms. Experimentally, three subband signals were generated with RLMD and VMD algorithms. It has been observed that the classification success of the designed deep learning algorithm decreases when fewer than three or more subband signals are used. The classification study was carried out on each channel of the 19-channel EEG signal, and each channel was applied to the deep learning algorithm separately and using various combinations.

The classification success rates obtained before the RLMD and VMD operations were compared with the classification success rates obtained at the output of the deep learning algorithm after the subband signals were obtained with RLMD and VMD.

It has been determined whether RLMD and VMD effectively detect the ADHD separately on the EEG signal, or not. It is discussed which method is more effective and the effect of applying RLMD and VMD together on the classification success achieved at the exit.

In this context, our study has unique characteristics. As seen in the literature summary, there are machine learning approaches and deep learning approaches where EEG signals are used in diagnosing ADHD.

As stated in the literature summary, it is seen that classification successes reaching 90% levels have been achieved with these approaches. The success rates obtained in our study are presented in section 3. With the developed method, high success rates have been obtained that will contribute to the literature on the diagnosis of ADHD.

In addition, the contribution of the RLMD and VMD approaches to the diagnosis of ADHD was specifically examined in our study. Using the contribution of RLMD and VMD, the findings of a highly successful classification study with minimum EEG channels are presented. The general flow of the study is shown in figure 1.

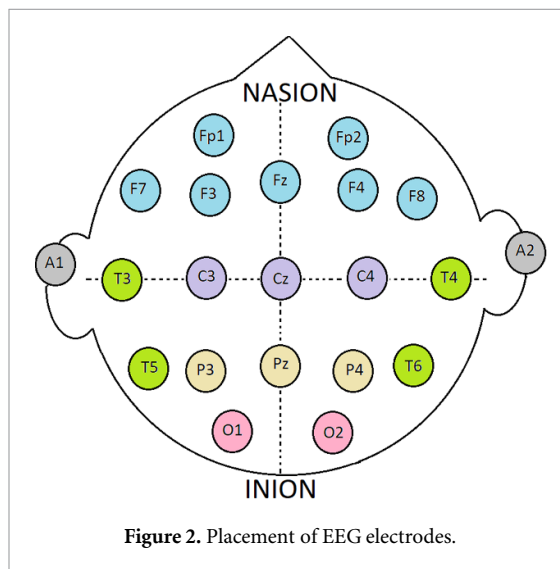
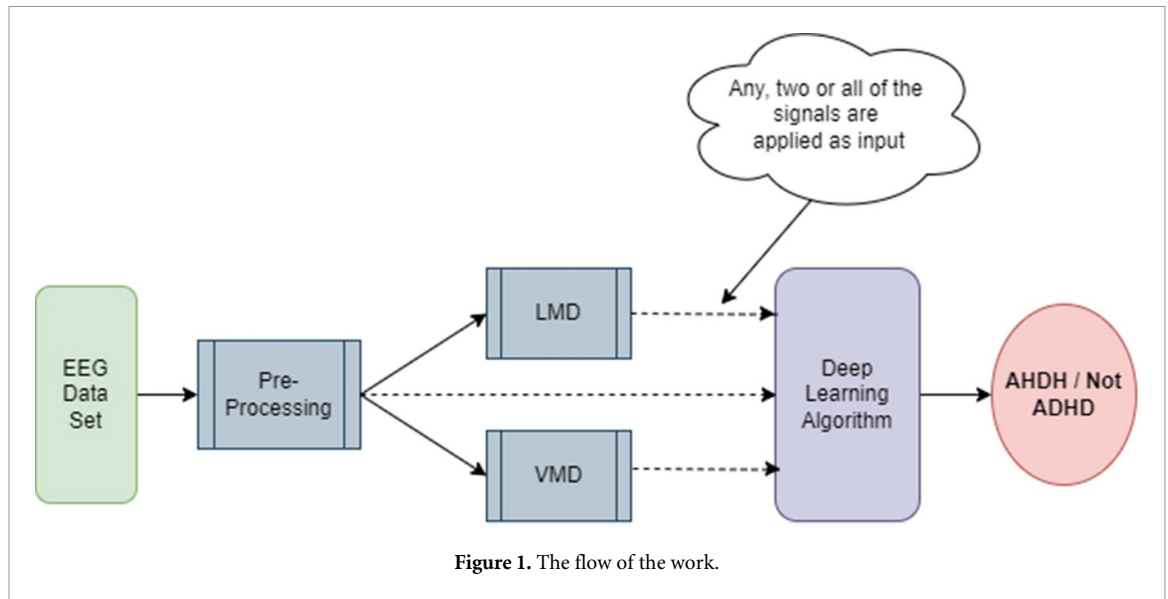
2.1. EEG data

The EEG data used in this study was obtained from the IEEE-DataPort database, as presented by Motie Nasrabadi *et al* [24]. While the EEG data was obtained from children aged 7–12 years, 61 of them had ADHD and 60 belonged to the healthy group-making a total of 121 participants. EEG signals were recorded as 19 channels according to the 10–20 standard based on the scalp locations: 'Fz, Cz, Pz, C3, T3, C4, T4, Fp1, Fp2, F3, F4, F7, F8, P3, P4, T5, T6, O1, O2', as shown in figure 2. A1 and A2 represent the reference electrodes which are attached to the earlobes. The sampling frequency of the recorded EEG signal was adjusted as 128 Hz. The data consisted of EEG recordings obtained while watching a visual stimulus. Details can be accessed from the specified database. As mentioned above, the placement of the EEG electrodes was based on the 10–20 system, as shown in figure 2.

Experienced psychiatrists diagnosed children with ADHD according to DSM-V criteria. Children with the ADHD had used Ritalin for up to six months. It was stated that none of the children in the control group had a history of psychiatric disorder, epilepsy or high-risk behaviour.

Since one of the deficiencies of individuals with ADHD is visual attention, it was stated that the EEG recording procedure took place during the visual attention task. In the task, children were shown pictures of cartoon characters and asked to count the characters. The number of characters in each picture was randomly chosen between 5 and 16, and the sizes of the pictures were chosen large enough to be easily seen and counted by the children. In order to have a continuous stimulus during signal recording, the following image continued display uninterruptedly, after the child's reaction to each image. Therefore, the duration of EEG recording during this cognitive visual task varied depending on the child's performance.

CNN structure has a superior ability to recognise patterns in an image. Therefore, our study applied EEG data to the CNN structure as an image matrix. EEG recordings in the database are of different lengths. A common data length was chosen to make a standard evaluation. In order to standardise EEG data of different lengths, data sets below a certain length were not included in the study. For this purpose, EEG signals with a data size of 9250 and above were included in the study. The data length was kept constant at 9250 as a standard to ensure the maximum number of participants and to work with the maximum EEG data. A large number of participants and an extended EEG dataset will create more training and test sets for the deep learning algorithm. This is important in terms of realistic evaluation of the algorithm's performance. EEG recordings of approximately 72 s were used. In this case, a total of 120 people, 61 with ADHD and 59 healthy,



$$\min_{\{u_k\}, \{w_k\}} \left\{ \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) x u_k(t) \right] \times e^{-j\omega_k t} \right\|_2^2 \right\}. \quad (1)$$

Here $\{u_k\} := \{u_1, \dots, u_K\}$ and $\{w_k\} := \{w_1, \dots, w_K\}$ are shorthand designations for the set of all modes and their centre frequencies, respectively. Likewise, $\sum_k := \sum_{k=1}^K$ is the sum of all modes [27].

In this study, subband signals were obtained by applying VMD to the EEG signal. Experimentally, when the number of subband signals is selected as 3, the highest learning success is achieved at the output of the deep learning algorithm. For this reason, the EEG signal was divided into three subbands by applying VMD.

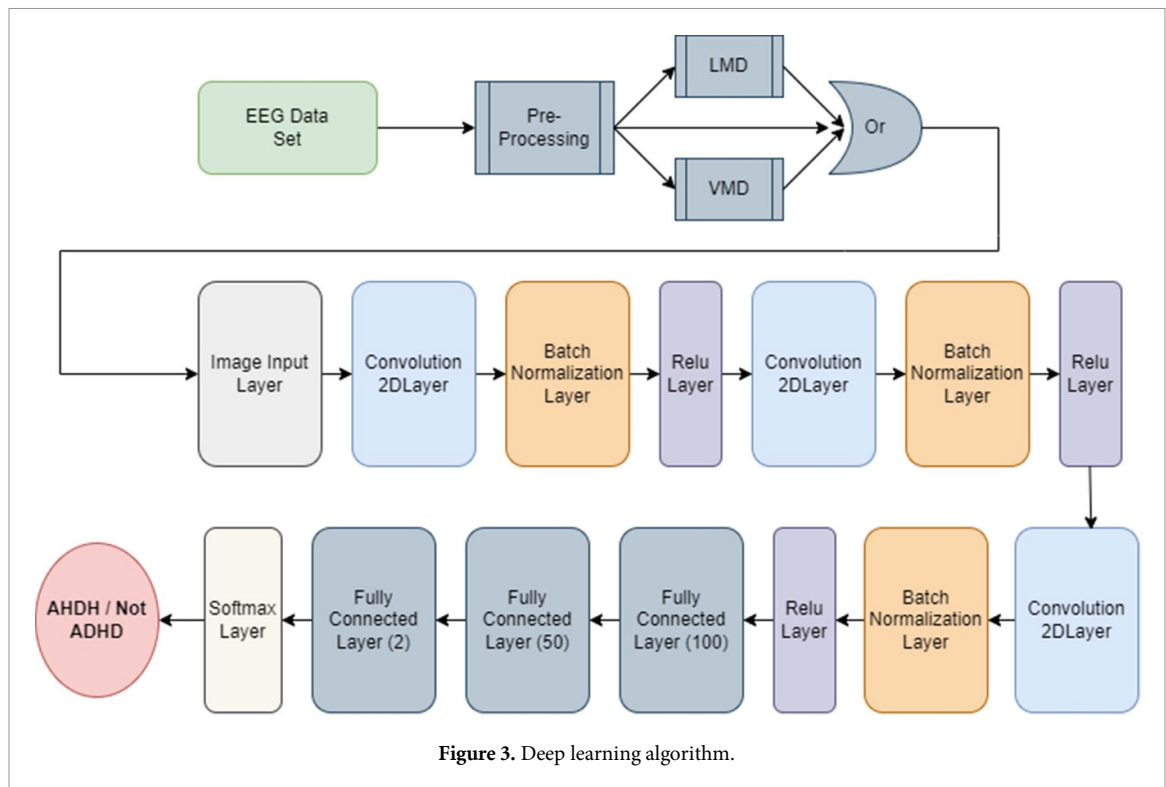
were included in the study. Only one person (from the healthy group) was not included in the study because the EEG recording length was too short. No selection was made among the participants. The entire data set (except one person specified) was included in the study.

2.2. VMD

VMD is a nonrecursive parsing algorithm developed to process time-varying and non-stationary signals. VMD can divide time series into reproducible bandwidth-limited submodes [25]. The VMD can decompose a real-valued signal 'f' into a discrete number of sub-signals 'uk', each having certain sparsity properties of the bandwidth in the signal's spectrum. Each mode is compact around the centre frequency (wk). The bandwidth is estimated using the 'H1' Gaussian smoothness of the shifted signal VMD [26]

2.3. RLMD

LMD involves gradually separating a frequency-modulated signal from an amplitude-modulated envelope signal. This separation is achieved by flattening the original signal, subtracting the smoothed signal from the original signal, and then amplitude demodulating the result using an envelope estimate. The envelope estimation is obtained using the accelerated weighted moving average between consecutive extremes of the original signal. This separation process continues until the remaining signal does not contain oscillations. In summary, the LMD of a signal is obtained by progressively smoothing the signal using the moving average [28]. The mathematical modelling of the LMD process and steps can be found in [28, 29]. As mentioned before, this study obtained subband signals by applying RLMD to EEG signals. RLMD is an optimised local mean decomposition. Optimisation strategies deal with boundary conditions, envelope



estimation, and elimination stop criteria in LMD. It simultaneously extracts many single component signals (called product functions) and their associated demodulation signals (i.e. AM and FM signals) from a mixed signal. RLMD can be used for time-frequency analysis [30]. In our case, the RLMD MATLAB function was used [31]. The signal was separated into three experimentally determined subbands as in VMD.

2.4. Deep neural network

In this study, the CNN structure, one of the deep neural network architectures, was used.

CNN has groundbreaking results in pattern recognition. It has many uses, from image processing to voice recognition. The most beneficial aspect of CNNs is that they reduce the number of parameters in artificial neural networks (ANNs). This encourages the use of CNN to solve complex tasks that are not possible with classical ANNs [32]. CNN consists of neurons that optimize themselves through learning. The notable difference between CNN and ANN is primarily the high performance of CNNs in pattern recognition within images [33, 34].

CNN basically consists of three-layer types: convolutional layers, pooling layers, and fully connected layers.

CNN architecture basically works as follows: convolutional layers determine the neuron outputs connected to the local regions of the input by multiplying the weights applied to the input. ReLu (rectified linear unit) aims to apply an activation function like sigmoid to the activation output produced by the previous layer. Pooling layers downsamples across

the spatial dimension of the given input, reducing the number of parameters within this activation. To arrive at the classification result, fully connected layers attempt to generate class scores from activations. It is recommended to use ReLu between layers to improve performance [32–34].

The CNN structure designed in our study was applied to EEG signals. EEG signals were applied as input data to the CNN structure, such as an image matrix, and the classification study was carried out.

2.5. Designed deep learning algorithm

Data processing and deep learning algorithm design was carried out using MATLAB version R2021b. First, each channel of the EEG signal is preprocessed, and the first 9250 data points of the signal are separated. Then, the subbands of the signals were calculated by performing RLMD and VMD applications. The signal and its subbands are applied as input data to the deep learning algorithm. The workflow of the algorithm is shown in figure 3.

After preprocessing, the EEG signal is applied to the algorithm as an input as if it were an image matrix. Then, convolution layer, batch normalization layer and ReLU layer operations are applied. This process is repeated three times. Then, it is tried to reach the classification result from the activation outputs obtained as 100-50-2 with three fully connected layer operations. Finally, the deep learning algorithm is completed with the softmax layer.

In the algorithm, the maximum epoch was chosen as eight, experimentally. In the lower selected epoch number, the classification success decreases; if it is



selected higher, the algorithm time becomes longer and does not contribute to the classification success. The study was carried out in five-folds, and the algorithm's performance was determined with the cross validation approach.

The algorithm runs for eight epochs and produces one of the confusion matrices seen in figure 4 after the training is complete. This process is repeated for five-folds, and the confusion matrix is produced at the end of each fold. In addition, the total duration of the training process was calculated, and the algorithm performance was recorded. The study was performed on a computer with an 11th-generation i7 processor, 16 GB RAM, and a 4 GB graphics card.

3. Result and discussion

In this study, the EEG signal data of the participating individuals belonging to healthy or ADHD group was used. The EEG records consisted of 19 channels, each of these channels was separately subjected to deep learning algorithm as input. The most effective EEG channel was determined by separately analysing the classification success of each channel.

Later, EEG signals from the brain's frontal, parietal, occipital, temporal and central lobes were used as input to the deep learning algorithm. Thus, the most effective brain zone in detecting ADHD was determined.

Finally, all EEG signals (19 channels) were simultaneously fed to the deep learning algorithm and the classification success achieved, when all channels are applied simultaneously, was determined.

As seen in figure 4, the five-fold algorithm was run, and success percentages were determined. Classification success was determined by taking the average of the success percentages obtained in each fold. Success rate, specificity, sensitivity and standard deviation parameters were calculated using the data obtained from the confusion matrix. Tables containing the findings of the whole study are presented in the continuation of the study.

The findings of the procedures expressed are shown in the tables and figures.

As seen in figures 5 and 6, if all 19 channel EEG signals were used in the classification study, the distinction between patient and healthy could be detected with 95.24% accuracy, 97% sensitivity, 94% specificity, and 1.78% standard deviation. The training period of the 19-channel study was determined as 1421 s in total. Such high success rate and short training duration of the algorithm is unprecedented and thought to enrich the literature. Chen *et al* have reported a success rate of 94% by using CNN for diagnosis of ADHD [17].

Consistent with the literature, the highest classification success was determined in the brain's frontal region. The frontal region contains seven-channel

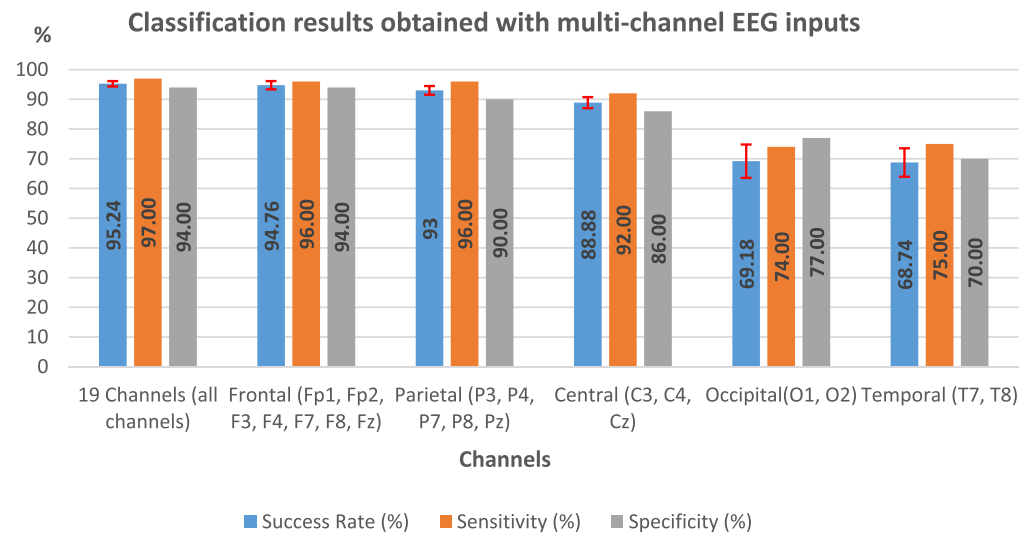


Figure 5. Classification results obtained with multi-channel EEG inputs.

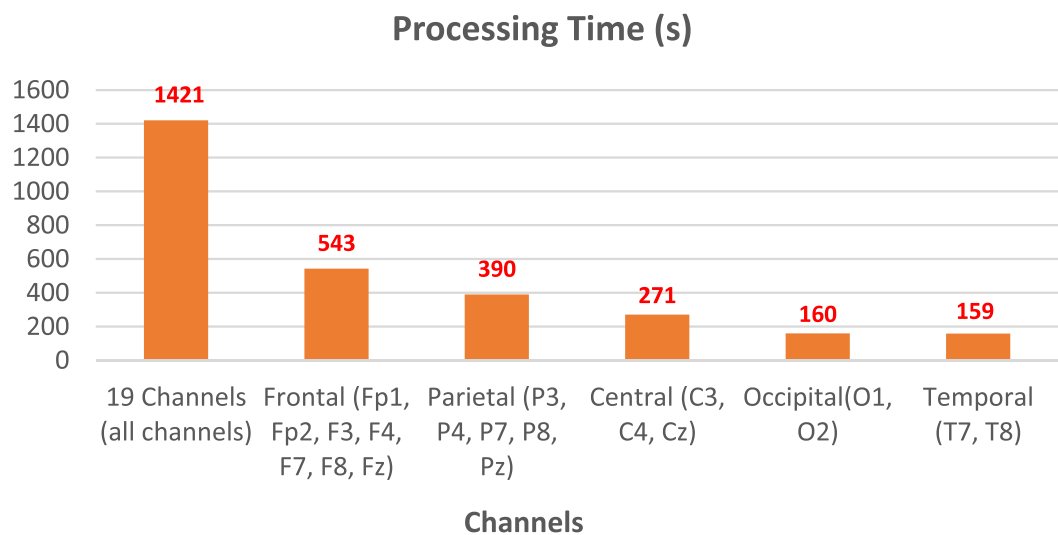
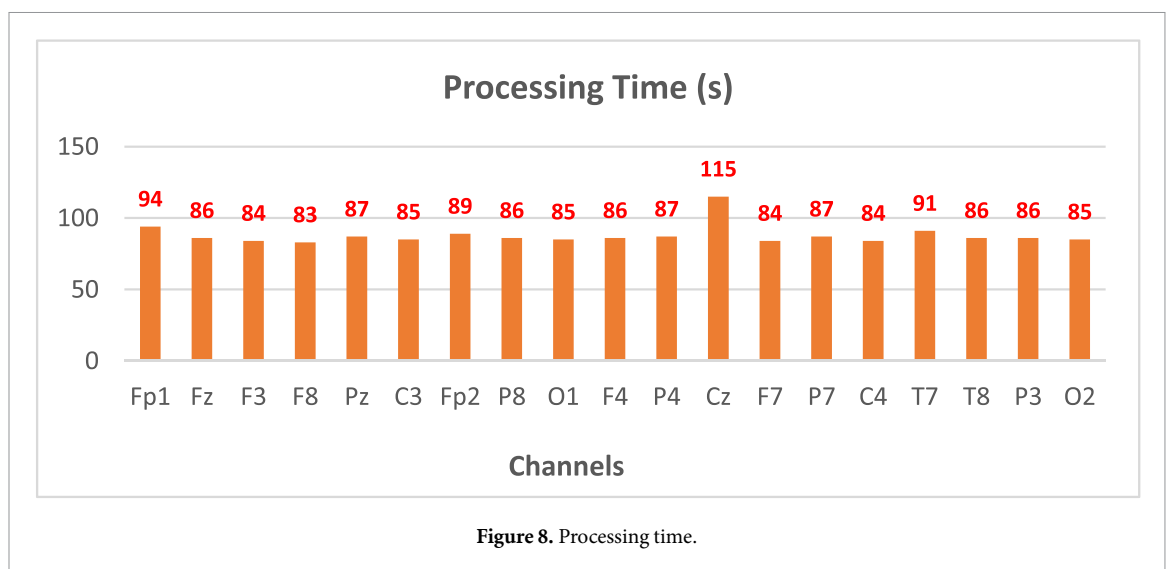
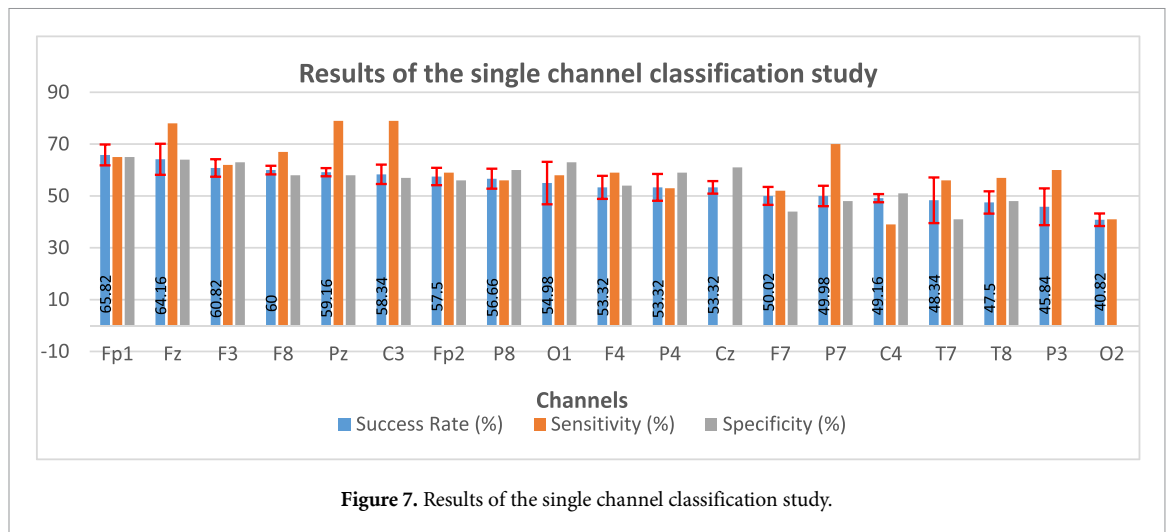


Figure 6. Processing time.

EEG recordings in parentheses in the figure 5. The result of the study, which was carried out only with the data obtained from the frontal region, could be determined with 95.76% accuracy, 96% sensitivity, 94% specificity, and 2.81% standard deviation. The training period of the study, which was carried out with seven-channel EEG signals from the frontal region, was determined as 543 s in total. Moghaddari *et al* has reported a classification success rate above 90% by using the dataset used in the current study with the deep neural networks [10, 20]. Information about the classification performance of other parts of the brain can also be noted in the figures 5 and 6.

It was tried to determine the most effective EEG channel in detecting ADHD. Each of the 19 channels was applied as input data to the deep learning algorithm in this context. In figure 7, the findings obtained as a result of the deep learning algorithm with single-channel signal input are presented. Results are listed based on the increasing success rate for each channel.

It can be seen in figures 7 and 8 that although the success rate acquired from the individual channels are lower, the algorithm times are shorter. The classification success of the channels in the frontal region (Fp1 and Fz) was generally higher



than the other channels. Ekhlasi *et al* also reported that the frontal channels are most effective in the diagnosis of the ADHD using EEG signals [23]. RLMD and VMD were applied to the channel with the highest classification success with the purpose of increasing the classification success using fewer channels and compensating the algorithm durations.

As mentioned above, although high success rates were obtained using many channels in the algorithm, the issue of longer durations persisted. Additionally, multi-channel recording is a challenging process in the clinic. Instead of EEG electrode connections covering the entire head surface, one-channel electrode connections will facilitate the clinical process. Tosun also reported that the Fp1 and F7 channels were most effective for diagnosis of the ADHD and a success rate of 88% was achieved by using only two channels [22].

This fundamental outcome of this study indicates that by sacrificing a small margin on the success rate using fewer EEG channels, clinical process-efficiency and EEG data collection-efficiency can be ensured. One step further, only highest success rate channels

obtained in the figure 7 were subjected to RLMD and VMD applications.

So, we tried to determine the most appropriate number of VMD and RLMD subbands for the EEG data set. When the EEG data set was divided into more than three subbands with VMD and RLMD, it was observed that some data were unsuitable, and more than three subbands could not be created. It was observed that the study using data suitable for creating more than three subbands did not improve the results. For this reason, it was deemed appropriate to experimentally divide the EEG data set into three subbands in VMD and RLMD applications.

Subband signals and the signal itself are applied as input data to the deep learning algorithm. The findings of the classification study are presented in table 1.

The findings presented in table 1 are explained in order below.

- In the application made in the 1st line; three sub-band signals were obtained from each application by applying RLMD and VMD to the Fp1 channel. Together with Fp1, seven signals were applied to

Table 1. Findings obtained as a result of RLMD and VMD application.

Row no	Channel and applications	Success rate (%)	Sensitivity	Specificity	Time (s)	Standard deviation (%)
1	Fp1 + rLmd + Vmd	87.38	95%	83%	646	1.99
2	Fp1 + rLmd	80.00	82%	82%	935	4.85
3	Fp1 → rLmd	66.94	71%	65%	818	6.50
4	Fp1 + Vmd	77.08	79%	80%	319	4.21
5	Fp1 → Vmd	63.34	62%	67%	248	4.30
6	Fp1	65.82	65%	65%	94	8.07

the deep learning algorithm (Fp1 + 3 subband VMD + 3 subband RLMD). About 87.38% classification accuracy was achieved.

- In the application made in the 2nd line; three subband signals were obtained from the application by applying RLMD to the Fp1 channel. Together with Fp1, four signals were applied to the deep learning algorithm (Fp1 + 3 subband RLMD). About 80% classification accuracy was achieved.
- In the application made on the 3rd line; unlike the implementation in line 2, only subband signals are used. About 66.94% classification accuracy was achieved.
- In the application made on the 4th line; three subband signals were obtained from the application by applying VMD to the Fp1 channel. Together with Fp1, four signals were applied to the deep learning algorithm (Fp1 + 3 subband VMD). About 77.08% classification accuracy was achieved.
- In the application made on the 5th line; unlike the implementation in line 4, only subband signals are used. About 63.34% classification accuracy was achieved.
- Only by applying Fp1 (without RLMD and VMD application) to the deep learning algorithm the results seen in the 6th line were obtained. About 65.82% classification accuracy was achieved.

In table 1, the findings obtained as a result of RLMD and VMD application to the Fp1 channel are presented. Findings are presented only for Fp1 because it is the channel with the highest classification success. As expected, lower success rates were obtained.

The highest success rate was achieved when the signal itself was applied to the deep learning algorithm along with the RLMD and VMD applications. A success rate of 87.38% was achieved by using a single channel. There has not been reported any study which attempts to diagnose ADHD using a single channel. This success is lower than that obtained from the 19 channels in figure 7. However, it is a much more easily applicable study in clinical practice. In addition, when the algorithm training times are compared, it is seen that the result is achieved in a much shorter time with a single channel. With the development of the algorithm, it is evaluated that more than 95% success can be achieved with a single channel by using VMD and RLMD.

As presented in table 1, lower success was achieved if the signal itself was not included in the VMD and RLMD applications. This situation shows that the signal has unique patterns that cannot be obtained in RLMD and VMD.

When RLMD and VMD applications are compared, it is seen that the contribution of RLMD to classification success is higher than VMD.

When RLMD and VMD were applied to the Fp1 signal, the ADHD classification success rate obtained at the algorithm output increased from 65.82% to 87.38%. This result emphasizes on the significance and effectiveness of this study.

It has been demonstrated that the deep learning algorithm, designed in this study, can detect ADHD with high success rates. Nineteen-channel EEG recording is a laborious process. Using single-channel EEG to detect ADHD will be an excellent convenience for the practising clinician and assisting staff. The proposed method is an original study for the detection of ADHD. The findings obtained with the proposed method are remarkable. As stated in the literature, it is considered that the findings obtained are not suitable for direct use in the clinic. However, with the development of the study, it is predicted that it can be used soon as a diagnostic support system that will help the clinician detect ADHD.

4. Conclusions

A summary of the data obtained from the literature is presented in table 2. Our study was analysed by comparing it with the literature, and the original aspects of our study were emphasised.

The findings obtained in the study are presented in figures and tables. As can be seen in figures 5 and 6, if the 19-channel EEG signal is applied to the deep learning algorithm, a success rate of over 95% has been achieved. A classification success rate of over 94% was achieved when only the frontal channels were applied. These results are compatible with the literature and contribute to it.

Among the 19 channels of the EEG signal, the channel with the highest contribution to ADHD detection was determined in figures 7 and 8. Using the results obtained from a single channel in diagnosing the disease will provide an excellent convenience for both the clinician and the patient/healthy participant because recording and evaluating a 19-channel EEG

Table 2. The overview of the previous literature on EEG and deep learning in ADHD.

Study	Data type	Method	Sample	Procedures and findings
Adamou <i>et al</i> [9]	EEG signal theta/beta ratio	Narrative analysis	21 Article (adult ADHD)	Theta/beta ratio. Not enough in adults to support the diagnosis
Arns <i>et al</i> [10]	EEG signal theta/beta ratio	Narrative analysis	6–18 years, 1253 with ADHD, 517 healthy	Not suitable as a diagnostic criterion. It can be used as a diagnostic support aid.
Gloss <i>et al</i> [11]	EEG signal theta/beta ratio	They classified them using American Academy of Neurology criteria	185 participants	EEG theta/beta power ratio and EEG frontal beta power to identify patients with ADHD correctly identified 166 of 185 participants
Tankoi <i>et al</i> [12]	EEG signal	EPSPatNet86 pattern learning network	The database used in our study [2]	EPSPatNet86 model can detect the ADHD. High accuracy up to 90% with EEG signal
Vahid <i>et al</i> [13]	EEG signal	‘EEGNet’ deep learning model	144 children (100 ADHD, 44 control)	ADHD subtypes and healthy controls with an accuracy of up to 83%.
Slater <i>et al</i> [14]	EEG signal	Review analysis	Can EEG identify ADHD subtypes? A systematic review	Although EEG signals are not yet sufficient to diagnose ADHD, they have great potential in general.
Quintan <i>et al</i> [15]	EEG signal	Statistical analyses	26 children/adolescent (16 ADHD, 10 control)	In the detection of ADHD in a diverse clinical sample, rating scales and EEG were both sensitive markers, whereas only EEG was specific.
Clarke <i>et al</i> [16]	EEG signal	Statistical analyses	25 adult man, 11 years apart	It has been stated that there are significant changes in the growth-related EEG and ADHD status.
Chen <i>et al</i> [17]	EEG signal	CNN, MLP, deep learning, classification	101 children (50 ADHD, 51 control)	The proposed framework achieved performance with an accuracy of 94.67% on the test data.
Dubreuil-Vall <i>et al</i> [18]	EEG signal	CNN, RNN deep learning, classification	40 adult (20 ADHD, 20 control)	The model classification accuracy of $88 \pm 1.12\%$
Chen <i>et al</i> [19]	EEG signal	CNN, deep learning	107 children (50 ADHD, 57 control)	The model classification accuracy of $90.29 \pm 0.58\%$
Moghaddari <i>et al</i> [20]	EEG signal	Signal processing, CNN, deep learning	61 children (31 ADHD, 30 control) The database used in our study [2]. Not all data were used in this study.	Classification success over 90% was achieved.
Tosun [22]	EEG signal	Machine learning, deep learning algorithms	The dataset consists of 272 samples. There are 17 recording states of 16 people per EEG channel.	It was stated that the accuracy of 88% (eyes open) was calculated in the ‘Fp1,F7’ channel. It was seen that the ‘Fp1,F7’ channel gave the highest success.

is an uncomfortable and tiring process for the clinician and patient/healthy participant. In addition, the registration and processing times are prolonged. The processing time of 19-channel EEG signals in the deep learning algorithm takes 1421 s.

In order to increase the classification success and to achieve maximum classification success with the minimum number of channels, RLMD and VMD, which are decomposition techniques, were applied

to the EEG signal. When we evaluate the findings presented in table 1,

- It has been determined that the effect of RLMD on classification success is higher than that of VMD.
- In the RLMD and VMD applications, the success obtained when using only subband signals is lower than the success obtained if subband signals are used together with the signal.

- A classification success rate of over 87% has been achieved when RLMD and VMD are applied to a single EEG channel (Fp1). This is the highest classification success achieved with a single channel in the literature. In addition, no classification study has been encountered in the literature for detecting ADHD with a single channel. In this context, our study adds to the literature.
- When the times specified in table 1 are examined, highly successful results were obtained much shorter than the classification of 19-channel EEG signals.
- ADHD affects EEG signals. When the EEG signal is evaluated as an image matrix, it has been determined that the pattern associated with ADHD on the matrix can be identified with high success with CNN.

According to the findings obtained from this study, results that will reduce the subjective effect of the clinician in the detection of ADHD were revealed by using a single-channel EEG signal. The importance of the EEG signal obtained from the Fp1 point of the frontal lobe for ADHD has been determined. It has been shown that using RLMD and VMD improves classification performance. Based on the success percentages, sensitivity, specificity, and standard deviation, it can be concluded that the study produced reliable results. It is predicted that with the development of the deep learning algorithm designed in the future, results with much higher classification accuracy will be obtained, and these results can be used as an aid to the clinician.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

Conflict of interest

None of the authors have potential conflicts of interest to disclose.

ORCID iDs

Mustafa Yasin Esas  <https://orcid.org/0000-0002-2677-5897>

Fatma Latifoğlu  <https://orcid.org/0000-0003-2018-9616>

References

- [1] Edition F 2013 Diagnostic and statistical manual of mental disorders *Am. Psychiatric Assoc.* **21** 59–74
- [2] Barkley R A 2006 *Attention-Deficit Hyperactivity Disorder: A Handbook for Diagnosis and Treatment* 3rd edn (New York: Guilford Press)
- [3] Spencer T J, Biederman J and Mick E 2007 Attention-deficit/hyperactivity disorder: diagnosis, lifespan, comorbidities, and neurobiology *Ambul. Pediatr.* **7** 73–81
- [4] Kieling R and Rohde L A 2010 ADHD in children and adults: diagnosis and prognosis *Behavioral Neuroscience of Attention Deficit Hyperactivity Disorder and Its Treatment Current Topics in Behavioral Neurosciences* vol 9, ed C Stanford and R Tannock (Berlin: Springer) pp 1–16
- [5] Conners C K, Sitarenios G, Parker J D A and Epstein J N 1998 The revised Conners Parent Rating Scale (CPRS-R): factor structure, reliability, and criterion validity *J. Abnorm. Child Psychol.* **26** 257–68
- [6] Conners C K, Sitarenios G, Parker J D A and Epstein J N 1998 Revision and restandardization of the Conners Teacher Rating Scale (CTRS-R): factor structure, reliability, and criterion validity *J. Abnorm. Child Psychol.* **26** 279–91
- [7] Tzimourta K D, Tsilimbaris A, Tzioukalia K, Tzallas A T, Tsipouras M G, Astrakas L G and Giannakeas N 2018 EEG-based automatic sleep stage classification *Biomed. J.* **7** 1–6
- [8] Craik A, He Y and Contreras-Vidal J L 2019 Deep learning for electroencephalogram (EEG) classification tasks: a review *J. Neural Eng.* **16** 031001
- [9] Adamou M, Fullen T and Jones S L 2020 EEG for diagnosis of adult ADHD: a systematic review with narrative analysis *Front. Psychiatry* **11** 871
- [10] Arns M, Conners C K and Kraemer H C 2013 A decade of EEG theta/beta ratio research in ADHD: a meta-analysis *J. Atten. Disord.* **17** 374–83
- [11] Gloss D, Varma J K, Pringsheim T and Nuwer M R 2016 Practice advisory: the utility of EEG theta/beta power ratio in ADHD diagnosis: Report of the Guideline Development, Dissemination, and Implementation Subcommittee of the American Academy of Neurology *Neurology* **87** 2375–9
- [12] Tanko D, Barua P D, Dogan S, Tuncer T, Palmer E, Ciaccio E J and Acharya U R 2022 EPSPatNet86: eight-pointed star pattern learning network for detection ADHD disorder using EEG signals *Physiol. Meas.* **43** 035002
- [13] Vahid A, Bluschke A, Roessner V, Stober S and Beste C 2019 Deep learning based on event-related EEG differentiates children with ADHD from healthy controls *J. Clin. Med.* **8** 1055
- [14] Slater J, Joobar R, Koborsy B L, Mitchell S, Sahlas E and Palmer C 2022 Can electroencephalography (EEG) identify ADHD subtypes? A systematic review *medRxiv*
- [15] Quintana H, Snyder S M, Purnell W, Aponte C and Sita J 2007 Comparison of a standard psychiatric evaluation to rating scales and EEG in the differential diagnosis of attention-deficit/hyperactivity disorder *Psychiatry Res.* **152** 211–22
- [16] Clarke A R, Barry R J, Johnstone S J, McCarthy R and Selikowitz M 2019 EEG development in attention deficit hyperactivity disorder: from child to adult *Clin. Neurophysiol.* **130** 1256–62
- [17] Chen H, Song Y and Li X 2019 A deep learning framework for identifying children with ADHD using an EEG-based brain network *Neurocomputing* **356** 83–96
- [18] Dubreuil-Vall L, Ruffini G and Camprodon J A 2020 Deep learning convolutional neural networks discriminate adult ADHD from healthy individuals on the basis of event-related spectral EEG *Front. Neurosci.* **14** 251
- [19] Chen H, Song Y and Li X 2019 Use of deep learning to detect personalized spatial-frequency abnormalities in EEGs of children with ADHD *J. Neural Eng.* **16** 066046
- [20] Moghaddari M, Lighvan M Z and Danishvar S 2020 Diagnose ADHD disorder in children using convolutional neural network based on continuous mental task EEG *Comput. Methods Programs Biomed.* **197** 105738
- [21] Lenartowicz A and Loo S K 2014 Use of EEG to diagnose ADHD *Curr. Psychiatry Rep.* **16** 1–11

- [22] Tosun M 2021 Effects of spectral features of EEG signals recorded with different channels and recording statuses on ADHD classification with deep learning *Phys. Eng. Sci. Med.* **44** 693–702
- [23] Ekhlasi A, Nasrabadi A M and Mohammadi M R 2021 Direction of information flow between brain regions in ADHD and healthy children based on EEG by using directed phase transfer entropy *Cogn. Neurodyn.* **15** 975–86
- [24] Motie Nasrabadi A, Allahverdy A, Samavati M and Reza Mohammadi M 2020 EEG data for ADHD/control children *IEEE Dataport* **10.21227/rzfh-zn36**
- [25] Wang R, Li C, Fu W and Tang G 2019 Deep learning method based on gated recurrent unit and variational mode decomposition for short-term wind power interval prediction *IEEE Trans. Neural Netw. Learn. Syst.* **31** 3814–27
- [26] Wang Y, Markert R, Xiang J and Zheng W 2015 Research on variational mode decomposition and its application in detecting rub-impact fault of the rotor system *Mech. Syst. Signal Process.* **60** 243–51
- [27] Dragomiretskiy K and Zosso D 2013 Variational mode decomposition *IEEE Trans. Signal Process.* **62** 531–44
- [28] Smith J S 2005 The local mean decomposition and its application to EEG perception data *J. R. Soc. Interface* **2** 443–54
- [29] Park C, Looney D, Van Hulle M M and Mandic D P 2011 The complex local mean decomposition *Neurocomputing* **74** 867–75
- [30] Liu Z, Jin Y, Zuo M J and Feng Z 2017 Time-frequency representation based on robust local mean decomposition for multi-component AM-FM signal analysis *Mech. Syst. Signal Process.* **95** 468–87
- [31] Zhiliang L 2022 Robust local mean decomposition (RLMD) (MATLAB Central File Exchange) (available at: www.mathworks.com/matlabcentral/fileexchange/66935-robust-local-mean-decomposition-rlmd) (Accessed 23 August, 2022)
- [32] Albawi S, Mohammed T A and Al-Zawi S 2017 Understanding of a convolutional neural network *2017 Int. Conf. on Engineering and Technology (ICET)* (IEEE) pp 1–6
- [33] O'Shea K and Nash R 2015 An introduction to convolutional neural networks (arXiv:1511.08458)
- [34] LeCun Y, Bengio Y and Hinton G 2015 Deep learning *Nature* **521** 436–44